

YUGESHKARAN P

GenAI Engineer · GenAI & Agentic Systems · LangChain · CrewAI · RAG · PyTorch
yugeshkaran01@gmail.com · +91 73975 92742 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#) · Bangalore, India

PROFESSIONAL SUMMARY

GenAI Engineer with hands on experience in building and deploying production-grade agentic AI systems using Python, LangChain, CrewAI, and PyTorch. Hands-on expertise architecting multi-agent LLM workflows, RAG pipelines, and secure API backends with measurable production outcomes. Experienced in bridging AI prototypes to enterprise-grade deployments — implementing guardrails, rate limiting, JWT/OAuth authentication, Redis caching, and CORS security at the system level. Adept at co-building with technical teams and converting real-world AI integration challenges into scalable reusable solutions.

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, LangGraph, LangSmith, CrewAI, Savram, ReAct, multi-agent orchestration, MCP, tool-calling, Gemini
ML & Deep Learning: PyTorch, HuggingFace Transformers, NLP, RAG pipelines, embedding models, LLM inference optimization
Languages: Python, JavaScript (ES6+), TypeScript, SQL, C — data structures, algorithms, software design
Backend & Security: Flask, Node.js, Express.js — REST APIs, JWT/OAuth, rate limiting, CORS, Redis caching, guardrails
Cloud & DevOps: AWS (S3, Lambda, Amplify), Docker, Git/GitHub, CI/CD, Linux/Unix, Vercel
Databases & Vectors: MySQL, MongoDB, Redis, Pinecone VectorDB — semantic retrieval, schema design, query optimization

PROJECTS

- Assistant Knowledge Hub — Production RAG System** [Github](#) *Python · Flask · OpenAI · Pinecone VectorDB · LangChain · Pydantic*
- Built a production-grade RAG assistant with OpenAI text-embedding-3-small and Pinecone VectorDB, enforced **20 requests/minute/user rate limiting** to optimize LLM API costs and ensure equitable resource allocation across concurrent users.
 - Implemented **CORS middleware blocking 100% of cross-site request forgery** attempts and deployed **LLM guardrails preventing 100% of prompt injection attacks** — delivering enterprise-grade security for a production agentic deployment.
- Indic Research Scholar — Production RAG System** [Github](#) *Flask · Sarvam Models · Pinecone VectorDB · Node.js · MongoDB · React.js · TailwindCSS*
- Full-stack RAG platform helps researchers to query academic papers in 8+ Indian languages, with OCR ingestion and voice I/O via Sarvam AI (105B LLM, Saaras v3 STT, Bulbul v3 TTS).
 - Built end-to-end auth (JWT + GitHub OAuth), multi-tenant document isolation, and a synchronous.
- Mentor Consulting Crew AI — Multi-Agent System** [Github](#) *Python · CrewAI · Gemini 2.0 Flash · ScrapeWebsiteTool · SerperDevTool*
- Architected a production multi-agent system using CrewAI with hierarchical delegation and ReAct patterns, agents autonomously perform web research, scrape content sources, and generate structured curriculum plans with efficient task distribution across specialized roles.
 - Integrated Gemini 2.0 Flash as the core LLM backbone with ScrapeWebsiteTool and SerperDevTool for live data retrieval, designed agentic workflows with self-reflection loops to improve output accuracy and reduce hallucination in research synthesis tasks.
- Mini Language Model — Custom NLP Training** [Github](#) *Python · PyTorch · Transformers · Flask*
- Designed and trained a 2.96M-parameter transformer-based LLM from scratch using PyTorch.
 - Built an end-to-end pipeline covering tokenization, architecture design, optimization, and inference speed and memory efficiency tuning and exposed via Flask inference API.
- AI Assisted Tech Community Platform** [Github](#) *Node.js · Flask · React.js · MongoDB · Redis · AWS S3 · Socket.io · JWT*
- Implemented **Redis server-side caching** on content feed API achieving **80% reduction in API calls** per user session, replaced session-based auth with **JWT authentication** across all client requests, eliminating session-store performance bottlenecks.
 - Engineered API-level **rate limiting middleware** preventing **100% of abusive requests and DoS attack vectors**, built real-time Socket.io discussions, AWS S3 media storage, and deployed on Vercel with CI/CD practices.
 - Built responsive React.js/TailwindCSS frontend and Framer motion, deployed on Vercel with CI/CD practices and maintained high availability under concurrent user load.

INTERNSHIP EXPERIENCE

- Full-Stack Web Development Intern · Quantum Sharq Innovative Solutions** 2024
- Built and optimized React.js/TailwindCSS web applications for clients, developed modular HRMS components in an Agile team, maintaining code quality through reviews and testing.
- Data Science Intern · Comp Allah IT Industry** 2023
- Performed data analysis, ML model training, and Python diagnostic scripting on business datasets, built visualizations to drive data-informed decision-making workflows.

EDUCATION

B.Tech — Artificial Intelligence & Data Science · Dhanalakshmi Srinivasan University, Trichy 2021 – 2025
CGPA: **9.21 / 10.0** · Class XII (CBSE): 92%

RESEARCH PUBLICATIONS

- (Jun 2025)** "Evaluating the Performance of LLMs on SQL and NoSQL Databases Using LangSmith" — Intl. Conf. Multidisciplinary Solutions for Emerging Trends 2025
- (Sep 2024)** "Design and Development of an Attendance Management and Analysis System Using LLM" — ICIAAIEA 2024